Aylin Davoudi

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PROFILE

Aerospace engineering master's student with a multidisciplinary background across diverse engineering projects, specializing in space propulsion. Seeking a 4-6 month end-of-studies internship starting in March/April 2027.

EDUCATION

Master of Aerospace Engineering – ISAE-SUPAERO, Toulouse, France
Advanced Aerodynamics & Propulsion Major - Space Track

Relevant Coursework: *Space Propulsion, Space Systems, Satellite Thermal Control Systems, Space Environment, Space Law, Computational & Experimental Fluid Dynamics, Satellite Propulsion: Chemical and Electrical, Launchers Architecture, Multiphase Flow and Combustion, Physics and Modeling Turbulence, Acoustics, Orbital Mechanics, Embedded Systems, Electromagnetism applied to Avionics, Flight Dynamics, High Performance Computing, Control Engineering,* *2025–2027*

International Bachelor in Mechanical, Materials & Aerospace Engineering – INSA Lyon

Relevant Coursework: *Fluid Mechanics, Heat Treatments, Material Science & Structural Analysis, Composite Materials, Machine Elements, Transducers & Measurements, Flight Dynamics & Control, Mechanical Design, Manufacturing* *2022–2025*

Courses at ECAM LaSalle (partnership with IBMMAE, INSA Lyon):
Compressible Flows & Propulsion, Fluid Mechanics, Thermodynamics, Manufacturing & CNC Machining) *2022–2025*

Exchange Semester – University of Twente, The Netherlands:*Fluid Mechanics 2, Heat Transfer, Control and Mechatronics* *FEB– JUN 2024*

High School Diploma – Georgian American International High School, Tbilisi, Georgia *2020–2021*

WORK EXPERIENCE

Thermal Cooling Team Leader — PERSEUS (CNES) / ISAE SUPAERO *2026 – present*

- Designing regenerative cooling circuits for 15 kN LOX–ethanol rocket engines.

Engineering Intern — CMD Gears, Cambrai, France *Apr 2025 – Aug 2025*

- Modeled conical interference fits to evaluate contact pressure, material behavior, and torque transmission (FEMAP, Solid Edge, KISSsoft/KISSsys).
- Investigated lubrication effects and conducted analytical/FEM parametric studies and evaluated them against experimental studies carried out on the factory test-bench; contributed to bearing design, gear meshing, and structural analyses.
- Delivered technical reports and engineering justifications for internal and client projects.

Assembly Line Worker — Stellantis, Poissy, France *Jun – Aug 2023*

- Performed assembly operations (kitting, leak testing, wheel and bumper assembly) on automotive production lines.
- Developed precision, adaptability, and teamwork in a fast-paced manufacturing environment.

PROJECTS

Droplet Evaporation in Real-Gas & Supercritical Conditions – Master's Research Project *ISAE-SUPAERO* *2025–Ongoing*

- Extended a Python-based evaporation solver (D^2-law, Abramzon–Sirignano) to real-gas conditions by implementing a Peng–Robinson equation of state with multi-component mixing rules.
- Applied fugacity-based corrections to thermodynamic properties to model high-pressure environments.
- Currently working on extending the solver to supercritical conditions via a diffuse-interface (Langmuir–Knudsen) treatment coupled with Helmholtz free-energy EOS.

Design & Optimization of a Pintle Injector – Bachelor Thesis *INSA Lyon* *2024–2025*

- Designed and modeled a RP-1/LOX based pintle injector for lander propulsion, developing analytical models for mass flow, momentum ratio, and combustion performance.
- Performed CFD simulations (Star-CCM+) to evaluate the influence of pintle angle and gap on mixing efficiency and thrust.
- Built and tested a physical prototype, validating results against analytical predictions and CFD simulations.

Design & Optimization of Regenerative Cooling Channels for Vulcain 2 – *University of Twente* *2024*

- Designed and optimized regenerative cooling channels for a rocket nozzle using analytical methods and CFD to maintain safe wall temperatures.
- Analyzed flow and heat transfer to refine channel geometry and improve cooling efficiency.
- Validated performance through wind tunnel experiments, comparing heat transfer measurements with theoretical predictions.

Mechatronic Design of a Mirror Manipulator for Inter-Satellite FSO Communication – *University of Twente* *2024*

- Designed a precision mirror positioning system, defining tracking requirements and implementing a PID controller with feedforward control.
- Modeled and simulated system dynamics (Spacar, Simulink); designed compliant flexures in SolidWorks with safety validation; built and tested the prototype.
- Achieved lowest tracking error among all groups ($< 30\,\mu$rad).

Ultrasound-Based NDT for Aircraft Maintenance – *INSA Lyon* *2024*

- Designed and simulated ultrasonic sensors in COMSOL for defect detection in aerospace structures, optimizing piezoelectric configurations and placement.
- Analyzed wave propagation in composite materials to improve detection of cracks, delamination, and voids for predictive maintenance.

NACA 0012 Airfoil Optimization: Experimental & Computational Aerodynamics – *ISAE-SUPAERO* *2025–2026*

- Ran wind-tunnel experiments on a NACA 0012 airfoil, testing Gurney flaps and boundary-layer trips via force balance and pressure-tap measurements
- Built and validated 2D RANS CFD simulations (STAR-CCM+, SST k-omega $\lambda - Re\theta$) against experimental data
- Identified optimal Gurney flap sizing for best lift-to-drag trade-off; quantified CFD limitations in predicting stall at low Reynolds numbers.

Fabrication, Characterization & Simulation of Unidirectional Composites – *INSA Lyon* *2024*

- Fabricated composite samples using contact molding and vacuum techniques; performed mechanical characterization (tensile tests, DIC, vibratory analysis).
- Simulated mechanical behavior in Abaqus to analyze stress response and optimize stacking sequences.

SKILLS

Programming	Python, C/C++, HTML, CSS, JavaScript, Fortran
Software	MATLAB/Simulink, SolidWorks, Solid Edge, Fusion 360, Ansys, Star-CCM+, COMSOL Multiphysics, Abaqus, FEMAP, KISSsoft/KISSsys, XFOIL/XFLR5, OpenVSP, IBM Engineering Rhapsody (SysML), SU2, GetFEM++, Gmsh, ParaView, Microsoft Office, GitHub/GitLab, ISAE softwares: Longisim, Transsim. Timeus
Machining	Certified HAAS lathe and mill operator, G-code programming
Languages	English (IELTS Academic 8.5), French (intermediate), Spanish (intermediate), Portuguese (beginner)